Assume/Guarantee Contract of `GetGyroData`

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1 Variable Declaration

Maps : `variables` in Code $\mapsto$ variables in Formal Specification

Alphabet :
wa′, ADDR_AD_START, ADDR_AD_START′, AD_GYRO_LO4, AD_GYRO_LO4′, AD_GYRO_HI8, AD_GYRO_HI8′, chnl, chnl′, type, type′, clock, clock′

Inputs : None

Outputs :

- `wa` $\mapsto$ wa : $\mathbb{N} \rightharpoonup \mathbb{R}$

In-Outputs : None

States : None

Parameters :

- `ADDR_AD_START` $\mapsto$ ADDR_AD_START : $\mathbb{N}$
- `AD_GYRO_LO4` $\mapsto$ AD_GYRO_LO4 : $\mathbb{N}$
- `AD_GYRO_HI8` $\mapsto$ AD_GYRO_HI8 : $\mathbb{N}$
- `chnl` $\mapsto$ chnl : $\mathbb{N} \rightharpoonup \mathbb{N}$
- `type` $\mapsto$ type : $\mathbb{N} \rightharpoonup \mathbb{N}$

Clocks : clock, clock′ : $\mathbb{R}_{\geq 0}$

2 AG Contract

Assumptions :

$$Assumptions \triangleq \mathsf{len}(\mathsf{chnl}) = \mathsf{len}(\mathsf{type}) = 9$$

Guarantees :

$$\begin{aligned}
Guarantees &\triangleq Check_1 \wedge Check_2 \\
Check_1 &\triangleq \mathsf{ADDR_AD_START}' = \mathsf{ADDR_AD_START} \\
&\quad \wedge \mathsf{AD_GYRO_LO4}' = \mathsf{AD_GYRO_LO4} \\
&\quad \wedge \mathsf{AD_GYRO_HI8}' = \mathsf{AD_GYRO_HI8} \\
&\quad \wedge \mathsf{chnl}' = \mathsf{chnl} \\
&\quad \wedge \mathsf{type}' = \mathsf{type} \\
&\quad \wedge \mathsf{clock}' = \mathsf{clock} + 5.4 \times 10^{-6} \\
Check_2 &\triangleq \forall 0 \leq i < 9 \cdot \exists \mathsf{wd} : \mathbb{N} \cdot \\
&\quad \left(\begin{array}{l} \mathsf{if}\ \mathsf{type}[i] = 0\ \mathsf{then} \\ \quad \mathsf{wa}'[i] = \mathsf{DiviationToFloat}(\mathsf{wd}, \mathtt{0xFFF}, \mathtt{0x800}, 0.00162760417) \\ \mathsf{else} \\ \quad \mathsf{wa}'[i] = \mathsf{DiviationToFloat}(\mathsf{wd}, \mathtt{0xFFF}, \mathtt{0x800}, 0.0009765625) \\ \mathsf{endif} \end{array} \right)
\end{aligned}$$